MILITARY SPECIFICATION

TUBES, TRACHEAL, MURPHY, Sterile, Plastic, Disposable

This limited coordination military specification has been prepared by the Defense Personnel Support Center based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-T-37133A. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

1. SCOPE

1.1 Scope. This specification covers sterile, plastic, disposable, Murphy-type tracheal tubes having an inflatable cuff with pilot balloon and mechanical inflation valve.

1.2 Classification. Tubes shall be furnished in accordance with the sizes specified in Table I (Tracheal Tube Dimensions) of American National Standards Institute (ANSI) Z79.14.

1.3 Style. Tubes shall be furnished in the styles indicated below (see 6.1):

Style A - Radiopaque
Style B - Nonradiopaque

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6515

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:02Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

MMM-A-250	-	Adhesive, Water-Resistant, (For Closure of Fiberboard Boxes).
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-B-676	-	Boxes, Setup.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z70.1	Dimensions of Glass and Metal Luer Tapers for Medical Applications.
ANSI Z79.14	Anesthetic Equipment - Tracheal Tubes.
ANSI Z79.16	Anesthetic Equipment - Cuffed Oral Tracheal and Nasal Tracheal Tubes for Prolonged Use.
ANSI Z79.2	Tracheal Tube Connectors and Adapters.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 412	Standard Test Methods for Rubber Properties in Tension.
ASTM D 2240	Standard Test Method for Rubber Property-Durometer Hardness.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

U. S. PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States (USP)

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:02Z
Check the source to verify that this is the current version before use.

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material. The material indicated herein shall be of the highest quality and shall be suitable in all respects for the purpose intended.

3.1.1 Tracheal tube. The tracheal tube shall be fabricated of a polyvinyl chloride plastic and shall conform to the Pharmacopeia of the United States (USP) requirements for Class IV plastics. The tracheal tube shall be tested for biological safety as specified in 4.4.1.

3.1.2 Cuff. The cuff shall be fabricated of polyvinyl chloride plastic and shall conform to the USP requirements for Class IV plastics. The cuff shall be tested for biological safety as specified in 4.4.1.

3.1.3 Inflation tube and pilot balloon. The inflation tube and pilot balloon shall be fabricated of a polyvinyl chloride plastic and shall conform to the USP requirements for Class IV plastics. Shall be tested as specified in 4.4.1.

3.1.4 Inflation valve. The inflation valve shall be fabricated of a polyvinyl chloride and the poppet shall be fabricated from neoprene. If a valve spring is used, it shall be fabricated of AISI 300 Series corrosion-resistant steel.

3.1.5 Connector. The connector shall be fabricated of polypropylene and shall comply with the requirements of ANSI Z79.2.

3.2 Style and design. The tracheal tube shall consist of a permanently attached inflatable cuff, inflation tube, inflation valve, pilot balloon, and 15-millimeter (mm) connector. The tracheal tube shall be provided with a Murphy eye located at the patient end. The cuff, inflation system, and Murphy tips shall conform to the design requirements of ANSI Z79.14. Style I tracheal tubes shall have a radiopaque stripe extending the entire length of the tube. The luer fitting on the inflation valve shall be in accordance with ANSI Z70.1. The 15 mm tracheal tube straight connector shall be in accordance with ANSI Z79.2.

3.3 Dimensions. The dimensions and tolerances of the tracheal tube and the 15 mm connector and the luer fitting of the inflation valve shall be expressed in metrics in accordance with ANSI Z79.14.

3.4 Pressure test. The cuff and inflating system shall show no evidence of leakage when tested in accordance with 4.4.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:02Z
Check the source to verify that this is the current version before use.

3.5 Physical properties. The tracheal tubes shall have the following values for the respective tensile strength, elongation, and hardness when tested as specified in 4.4.4:

Size	Tensile strength (psi, min.)	Elongation (% min.)	Hardness Shore Durometer A (+5)
1	2300	120	85
2	2250	130	85
3	2000	130	85
4	2000	150	83
5	2000	150	83
6	2000	150	83
7	2000	150	83
8	1800	150	83
9	1750	150	83
10	1730	180	83

3.6 Lead content. The tracheal tube and cuff shall contain not more than five parts per million (ppm) of lead when tested as specified in 4.4.6.

3.7 Sterility. The tracheal tube shall be sterile when tested as specified in 4.4.7.

3.7.1 Certificate of sterility. A certificate of sterility, stating that samples from each sterilizer lot have been tested and found to be sterile, shall be submitted to the procuring activity. The certificate of sterility shall include:

a. Item identification.
b. National stock number (NSN).
c. Contract or purchase order number.
d. Sterility lot or control number.
e. Date of sterilization.
f. Quantity of product in sterilizer lot.
g. Type of sterilization (gas, radiation, heat, steam, etc.).
h. Form of biological indicator (i.e., inoculated paper spore strip or carrier, inoculated product, or inoculated simulated product).
i. Test organism(s) and population density used.
j. Test medium, incubation temperature, incubation time, and dates of tests.
k. Number of test units.
l. Statement that samples comply with sterility tests.
m. Signature of head of a laboratory or a designated qualified alternate.

When inspection is made at destination, the certificate of sterility shall be submitted by the contractor to the Directorate of Medical Materiel, ATTN: DPSC-ATQ, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, PA 19101, and a copy forwarded with each shipment to the consignee. When inspection is made at source, the certificate of sterility shall be furnished to the cognizant government quality assurance representative for submission to the procuring activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:02Z
Check the source to verify that this is the current version before use.

3.7.2 Sterility lot. A sterility lot is that single quantity or product subjected to the same manufacturing operation and sterilized simultaneously in the same sterilizer chamber. Each unit package shall be clearly marked with a sterility lot (control) number as specified in 5.3.1. Each sterility lot shall be traceable to the pertinent manufacturing lot(s).

3.7.3 Certificate of biological safety. A certificate of biological safety, listing all lot or control numbers, shall state that samples from each lot or control have been tested for compliance with the USP requirements for Class IV plastics as specified in 3.1.1, 3.1.2, 3.1.3 and 4.4.1 and were found to comply with such requirements.

3.8 Instructions. Each unit package shall be provided with instructions describing the proper techniques for dispensing the sterile tracheal tubes from the containers. In addition, information shall be provided describing the correct use of the inflation system.

3.9 Workmanship. The tracheal tubes (including components) shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificate of quality. A certificate of quality, supplied by the manufacturer of the plastic, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:02Z
Check the source to verify that this is the current version before use.

4.2 Sampling.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I.

TABLE I. Sampling for examination.

Type defect	Inspection level	AQL (percent defective)
Major	S-3	2.5
Minor	S-3	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics. The unit of product for test purposes shall be one tracheal tube.

TABLE II. Sampling for tests.

Component	Characteristic	Require-ment	Test procedure	Inspection level	AQL (percent defective)
Tracheal tube	Biological safety	3.1.1	4.4.1	S-1	1.5
Cuff	Biological safety	3.1.2	4.4.1	S-1	1.5
Inflation tube and pilot balloon	Biological safety	3.1.3	4.4.1	S-1	1.5
Inflation valve	---	3.1.4	---	--	--
Connector	---	3.1.5	---	--	--
Tracheal tube	Dimensions	3.3	4.4.2.1	S-2	4.0
Connector	Dimensions	3.3	4.4.2.2	S-2	4.0
Inflation valve	Dimensions	3.3	4.4.2.3	S-2	4.0
Tracheal tube	Pressure test	3.4	4.4.3	S-2	1.5
Tracheal tube	Tensile strength	3.5	4.4.4	S-1	1.5
Tracheal tube	Elongation	3.5	4.4.4	S-1	1.5
Tracheal tube	Hardness	3.5	4.4.5	S-1	1.5
Tracheal tube	Lead content	3.6	4.4.6	S-1	2.5
Cuff	Lead content	3.6	4.4.6	S-1	2.5
Tracheal tube	Sterility	3.7	4.4.7	*	*

* Acceptance number shall be zero.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:02Z
Check the source to verify that this is the current version before use.

4.3 Examination. The tracheal tubes shall be examined to determine compliance with specification requirements. Examination shall be conducted as specified in tables I and III. Examination, however, shall not be restricted to defcts listed in table III. Any tracheal tube in the sample containing one or more defects shall be rejected and, if the number of defective tubes in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

TABLE III. Classification of defects.

Categories	Defects
Major	
101	Tracheal tube not free from foreign matter.
102	Components missing or incomplete.
103	Depth markings incomplete, missing, illegible, or incorrect.
104	Tracheal tube not free from burrs, flashings, or other defect affecting usage, or, not free from roughness or defects which could be injurious to the patient.
105	Bevel edge at distal end of tracheal tube harder than rest of tube.
106	Pilot balloon excessively thick or stiff and not readily indicative of inflation.
107	Cuff not reasonably symmetrical when inflated as required in 3.4.
Minor	
201	Instructions illegible, inadequate, or missing.
202	Marking incorrect, incomplete, illegible, or omitted.

4.3.1 Packaging inspection. The inspection of the unit package, intermediate package, packing, and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Biological safety. Samples of the tracheal tube, cuff, and inflation tube and pilot balloon shall be tested in accordance with USP requirements for Class IV plastics.

4.4.2 Dimensions.

4.4.2.1 Tracheal tube. The dimensions and tolerances of the tracheal tubes shall be in accordance with ANSI Z79.14.

4.4.2.2 Connector. The dimensions and tolerances of the 15 mm connector shall be in accordance with ANSI Z79.2.

4.4.2.3 Inflation valve. The dimensions and tolerances of the Luer tip on the inflation valve shall be in accordance with ANSI Z70.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:02Z
Check the source to verify that this is the current version before use.

4.4.3 Pressure test. The cuff and the inflation system shall be inflated to an air pressure of 22 mm Hg and this pressure shall be maintained for 30 minutes.

4.4.4 Tensile strength and elongation. The tensile strength and elongation of the tracheal tube shall be determined in accordance with the American Society for Testing and Materials (ASTM) D 412.

4.4.5 Hardness. The hardness of the tracheal tube shall be determined in accordance with ASTM D 2240.

4.4.6 Test for lead. Ten grams shall be taken from the distal end of the tracheal tube or from the cuff, whichever is being tested, ignited to char, and then further ignited in a muffle furnace at 700° C to a white ash. The ash shall be boiled in 15 ml of 10% nitric acid, filtered into a volumetric flask and made up to 100 ml. An aliquot portion, equivalent to one gram of sample, shall be taken and placed in a separator. The dithizone test for lead, as given in the USP, shall be applied using a control of 5 ppm or 12 ppm of lead solution as given in the USP test. If the sample solution is darker than the limit, small aliquots shall be used until the color of the control is matched by the sample solution and the concentration of lead in ppm is determined. As an alternate test for lead, the following procedure may be followed:

Accurately weigh a 5-gram sample into a 1-ounce Vycor crucible. Reduce to ash in a muffle furnace at 650° C for eight hours. Let cool to room temperature and add 15 ml of 1:1 hydrochloric acid. Bring to a boil on a hot plate, allow to boil for 2 to 3 minutes, remove, let cool, and filter through Whatman #2 filter paper into a 50-ml volumetric flask. Rinse the residue on the filter with deionized water and dilute to 50 ml. Prepare standard lead solutions of 1, 2, and 5 ppm lead. Determine the absorbances of these solutions at the 283 millimicron wave length using a Hollow Cathode Pb lamp and 10X scale expansion on a spectrophotometer, such as the Perkin-Elmer Model 303 Atomic Absorption Spectrophotometer. Determine the absorbance of the sample solutions using the same conditions as with the standards.

Calculation:

$$\frac{A \times F \times \text{dilution}}{W} = \text{ppm lead wherein,}$$

A = Absorbance

F = Factor = $\frac{\text{(Concentration standard in ppm Pb)}}{\text{absorbance}}$ 1/

W = Sample weight in grams

1/ The three standards are averaged to give ppm/unit A.

4.4.7 Sterility test. The tracheal tube shall be tested to determine its sterility in accordance with the USP for Parenteral devices.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:02Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One package containing ten tubes of the size specified constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Individual package. Each tube shall be packaged in a sealed peel-open type bag capable of maintaining sterility of contents unless damaged or opened. The bag shall be designed with tabs to facilitate opening.

5.1.2.2 Unit package. Ten individual packages, containing tubes of one size only, shall be packaged in a box conforming to PPP-B-566 or PPP-B-676. Closure of the bag shall be as specified in the appendix of the applicable box specification.

5.1.2.3 Intermediate package. Five packages containing tubes of one size only shall be packed in a box conforming to PPP-B-636, type CF, class domestic. Closure shall be as specified in the box specification.

5.1.3 Level C. Ten tubes of one size only individually packaged as specified in 5.1.2.1 shall be further packaged in a manner that will afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Two intermediate packages, containing tubes of one size only, shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B. Two intermediate packages, containing tubes of one size only, shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged tubes shall be packed in shipping containers in a manner that will afford adequate protection against physical damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier's rules and regulations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:02Z
Check the source to verify that this is the current version before use.

5.3 Marking.

5.3.1 Individual package. In addition to any commmercial markings, each package shall be marked in accordance with the requirements of the Federal Food, Drug and Cosmetic Act. In addition, the bag shall be marked with the name and size of the commodity contained, the sterility lot (control) number, and the legends: "STERILE" and "STERILITY GUARANTEED UNLESS PACKAGE IS DAMAGED OR OPENED" or similar statement.

5.3.2 Unit package and intermediate package. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. The sterility lot (control) number shall be shown.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Sterility lot (control) number shall be shown.

6. NOTES

6.1 Intended use. The tracheal tubes are intended for use in nasal and oral intubation.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Style (see 1.3).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
d. Applicable NSN.

6.3 National stock number coverage. This specification covers the following items appearing in the Federal Supply Catalog:

NSN	Item Identification	Specification Designation Size	Style
6515-00-105-0653	Tube, Tracheal, Murphy, with Cuff, 3 mm ID, 10s	1	B
6515-00-105-0664	Tube, Tracheal, Murphy, with Cuff, 4 mm ID, 10s	2	B
6515-00-105-0707	Tube, Tracheal, Murphy, with Cuff, 5 mm ID, 10s	3	A
6515-00-105-0720	Tube, Tracheal, Murphy, with Cuff, 6 mm ID, 10s	4	A
6515-00-105-0744	Tube, Tracheal, Murphy, with Cuff, 7 mm ID, 10s	5	A

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:02Z
Check the source to verify that this is the current version before use.

NSN	Item Identification	Specification Designation	
		Size	Style
6515-00-105-0759	Tube, Tracheal, Murphy, with Cuff, 8 mm ID, 10s	6	A
6515-00-105-0808	Tube, Tracheal, Murphy, with Cuff, 9 mm ID, 10s	9	A
6515-00-105-0815	Tube, Tracheal, Murphy, with Cuff, 10.0 mm ID, 10s	10	A
6515-01-036-9034	Tube, Tracheal, Murphy, with Cuff, 7.5 mm ID, 10s	7	A
6515-01-036-9035	Tube, Tracheal, Murphy, with Cuff, 8.5 mm ID, 10s	8	A

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army-MD
Navy-MS
Air Force-03

Preparing activity:
DoD-MB

Agent:
DLA-DM

Project No. 6515-2612

☆U.S. GOVERNMENT PRINTING OFFICE: 1984—705-040/A1940

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:02Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEFENSE LOGISTICS AGENCY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 4966 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEFENSE LOGISTICS AGENCY

Defense Personnel Support Center
Directorate of Medical Materiel, Code ATT
2800 South 20th Street
Philadelphia PA 19101

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:02Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:02Z
Check the source to verify that this is the current version before use.